NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1513
Additional Practice Set 4 Solutions

1. $\det(\lambda I - A) = \begin{vmatrix} \lambda & -1 & 0 \\ 1 & \lambda-1 & -1 \\ 0 & -1 & \lambda \end{vmatrix} = \lambda^2(\lambda - 1)$

   So the eigenvalues are 0 (multiplicity 2) and 1.

   Eigenspace for $\lambda = 0$: $\left(\begin{array}{ccc|c} 0 & -1 & 0 & 0 \\ 1 & -1 & -1 & 0 \\ 0 & -1 & 0 & 0 \end{array}\right) \xrightarrow[R_3 - R_1]{R_2 \leftrightarrow R_1} \left(\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$.

   We have $z = t, y = 0, x = t$, so the eigenspace contains eigenvectors $t\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$.

   Eigenspace for $\lambda = 1$: $\left(\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 \end{array}\right) \xrightarrow[R_3 - R_2]{R_2 - R_1} \left(\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$.

   We have $z = t, y = t, x = t$, so the eigenspace contains eigenvectors $t\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

   (ii) $\boldsymbol{A}$ is not diagonalizable, since the eigenvalue 0 has multiplicity 2 but only correspond to one linearly independent eigenvector (or simply: we can only find two linearly independent eigenvectors for $\boldsymbol{A}$.)

   (iii) The system has infinitely many solutions.

   $\boldsymbol{A}$ has eigenvalue $0 \Rightarrow \boldsymbol{A}$ is singular $\Rightarrow \boldsymbol{A}^{100}$ is singular $\Rightarrow \boldsymbol{A}^{100}\boldsymbol{x} = \boldsymbol{0}$ has nontrivial solutions.

2. (i) Since $\boldsymbol{A}$ is a triangular matrix, the eigenvalues are given by the diagonal entries $1, 2, 3, 4$.

   (ii) Yes. Since A is 4×4 and it has 4 distinct eigenvalues. So $\boldsymbol{A}$ is diagonalizable.

   (iii) We can check by pre-multiply $\boldsymbol{A}$ to each of the given vectors:

   $$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 0 \\ 0 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

   So $\boldsymbol{v}_1$ is an eigenvector associated with eigenvalue 2.

   $$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 3 \\ 0 \end{pmatrix}.$$

So $\boldsymbol{v}_2$ is not an eigenvector.

$$\begin{pmatrix}1&1&1&1\\0&2&2&2\\0&0&3&3\\0&0&0&4\end{pmatrix}\begin{pmatrix}1\\1\\1\\1\end{pmatrix}=\begin{pmatrix}4\\6\\6\\4\end{pmatrix}.$$

So $\boldsymbol{v}_3$ is not an eigenvector.

$$\begin{pmatrix}1&1&1&1\\0&2&2&2\\0&0&3&3\\0&0&0&4\end{pmatrix}\begin{pmatrix}3\\4\\2\\0\end{pmatrix}=\begin{pmatrix}9\\12\\6\\0\end{pmatrix}=3\begin{pmatrix}3\\4\\2\\0\end{pmatrix}.$$

So $\boldsymbol{v}_4$ is an eigenvector associated with eigenvalue 3.

$$\begin{pmatrix}1&1&1&1\\0&2&2&2\\0&0&3&3\\0&0&0&4\end{pmatrix}\begin{pmatrix}8\\12\\9\\3\end{pmatrix}=\begin{pmatrix}32\\48\\36\\12\end{pmatrix}=4\begin{pmatrix}32\\48\\36\\12\end{pmatrix}.$$

So $\boldsymbol{v}_5$ is an eigenvector associated with eigenvalue 4.

(iv) To find $\boldsymbol{P}$, we need to find an eigenvector associated with eigenvalue 1.

We solve $(\boldsymbol{I}-\boldsymbol{A})\boldsymbol{x}=\boldsymbol{0}\Rightarrow\begin{pmatrix}0&-1&-1&-1\\0&-1&-2&-2\\0&0&-2&-3\\0&0&0&-3\end{pmatrix}\begin{pmatrix}x\\y\\z\\w\end{pmatrix}=\begin{pmatrix}0\\0\\0\\0\end{pmatrix}$

The general solution is $x=t, y=z=w=0$.

So an eigenvector associated with eigenvalue 1 can be given by $\begin{pmatrix}1\\0\\0\\0\end{pmatrix}$.

Then the matrix P can be $\begin{pmatrix}1&1&3&8\\0&1&4&12\\0&0&2&9\\0&0&0&3\end{pmatrix}$.

3. (i) $\boldsymbol{Au}=\begin{pmatrix}2&-1&1\\1&0&1\\1&-1&2\end{pmatrix}\begin{pmatrix}1\\1\\0\end{pmatrix}=\begin{pmatrix}1\\1\\0\end{pmatrix}$. So $\boldsymbol{u}$ is an eigenvector with eigenvalue 1.

$\boldsymbol{Av}=\begin{pmatrix}2&-1&1\\1&0&1\\1&-1&2\end{pmatrix}\begin{pmatrix}1\\2\\1\end{pmatrix}=\begin{pmatrix}1\\2\\1\end{pmatrix}$. So $\boldsymbol{v}$ is an eigenvector with eigenvalue 1.

$\boldsymbol{Aw}=\begin{pmatrix}2&-1&1\\1&0&1\\1&-1&2\end{pmatrix}\begin{pmatrix}2\\2\\2\end{pmatrix}=\begin{pmatrix}4\\4\\4\end{pmatrix}=2\begin{pmatrix}2\\2\\2\end{pmatrix}$. So $\boldsymbol{w}$ is an eigenvector with eigenvalue 2.

$\boldsymbol{Ax}=\begin{pmatrix}2&-1&1\\1&0&1\\1&-1&2\end{pmatrix}\begin{pmatrix}2\\1\\1\end{pmatrix}=\begin{pmatrix}4\\3\\3\end{pmatrix}$. So $\boldsymbol{x}$ is not an eigenvector.

(ii) From above, $\lambda = 1$ is an eigenvalue with multiplicity 2 and $\lambda = 2$ is an eigenvalue with multiplicity 1.
So the characteristic polynomial of $\boldsymbol{A}$ is $(x-1)^2(x-2)$.

(iii) From (i), the eigenspace for $\lambda = 1$ has a basis $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\}$;

and the eigenspace for $\lambda = 2$ has a basis $\left\{ \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} \right\}$.

(It is possible to use other vectors as the bases for the two eigenspaces.)

(iv) $\boldsymbol{P} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 0 & 1 & 2 \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.
(There are many possible answers for the matrix $\boldsymbol{P}$.)

(v) Eigenvalues of $4\boldsymbol{A}^{-1}$ are 4 and 2 (given by $4\lambda^{-1}$ where λ are eigenvalues of $\boldsymbol{A}$).

4. (i) From the characteristic polynomial $x^3 - 2x^2 + x = x(x-1)^2$, we get the eigenvalues 1 (repeated) and 0.
This means
the eigenspace of 1 is span $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$, and the eigenspace of 0 is span $\left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$.
So we can form
the matrix $\boldsymbol{P} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and the corresponding diagonal matrix $\boldsymbol{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

(ii) Suppose $\boldsymbol{v}$ is a vector that represent a point on the xyz-space that is mapped to the zero vector $\mathbf{0}$ under the linear transformation T.
$T(\boldsymbol{v}) = \mathbf{0} \Rightarrow \boldsymbol{A}\boldsymbol{v} = \mathbf{0}$.
Then $\boldsymbol{v}$ is an eigenvector of $\boldsymbol{A}$ corresponding to eigenvalue 0.
In other words, the subspace V is the eigenspace of 0, which is span $\left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$.

(iii) Note that the vector representing the populations in 2019:
$\begin{pmatrix} 400 \\ 300 \\ 100 \end{pmatrix} = 300 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + 100 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$. So it belongs to the eigenspace of 1, and hence is an eigenvector corresponding to eigenvalue 1.
This means $\boldsymbol{A} \begin{pmatrix} 400 \\ 300 \\ 100 \end{pmatrix} = \begin{pmatrix} 400 \\ 300 \\ 100 \end{pmatrix}$.

In particular,

$$\begin{pmatrix} a_n \\ b_n \\ c_n \end{pmatrix} = \boldsymbol{A}^n \begin{pmatrix} 400 \\ 300 \\ 100 \end{pmatrix} = \begin{pmatrix} 400 \\ 300 \\ 100 \end{pmatrix}$$

This means the three population groups remain the same every year.

5. (i) $\boldsymbol{M}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \boldsymbol{M}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \boldsymbol{M}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ -4 \end{pmatrix}.$

$\boldsymbol{M}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \boldsymbol{M}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \boldsymbol{M}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}.$

(ii) The three columns of $\boldsymbol{M}$ are given by $\boldsymbol{M}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \boldsymbol{M}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \boldsymbol{M}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$

So we have $\boldsymbol{M} = \begin{pmatrix} 0 & 2 & 0 \\ -2 & 2 & 0 \\ -4 & 2 & 2 \end{pmatrix}.$

(iii) Express the 3 columns of $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ as $\boldsymbol{M}\boldsymbol{v}_1, \quad \boldsymbol{M}\boldsymbol{v}_2, \quad \boldsymbol{M}\boldsymbol{v}_3$ for some vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$.

We have

$\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} = \boldsymbol{M}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$

$\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix} = \boldsymbol{M}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \boldsymbol{M}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \boldsymbol{M}\begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} = \boldsymbol{M}\left[\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}\right]$

$= \boldsymbol{M}\begin{pmatrix} -1 \\ 0 \\ -2 \end{pmatrix}.$

$\begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} = \boldsymbol{M}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$

So $\boldsymbol{M}\begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & 0 \\ 1 & -2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. Hence $\boldsymbol{X} = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & 0 \\ 1 & -2 & 1 \end{pmatrix}.$

(iv) $\frac{1}{2}\boldsymbol{M}\boldsymbol{X} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$

So $\boldsymbol{M}^{-1} = \frac{1}{2}\boldsymbol{X} = \frac{1}{2}\begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & 0 \\ 1 & -2 & 1 \end{pmatrix}.$

6. (i)

$$T\left(\begin{pmatrix}1\\0\\0\end{pmatrix}\right)=\begin{pmatrix}0\\1\\0\end{pmatrix},\quad T\left(\begin{pmatrix}0\\1\\0\end{pmatrix}\right)=\begin{pmatrix}-1\\0\\0\end{pmatrix},\quad T\left(\begin{pmatrix}0\\0\\1\end{pmatrix}\right)=\begin{pmatrix}0\\0\\-1\end{pmatrix}$$

(ii) $\boldsymbol{A}=\begin{pmatrix}0&-1&0\\1&0&0\\0&0&-1\end{pmatrix}$ (the three columns are the respective images in (i) above.)

(iii)

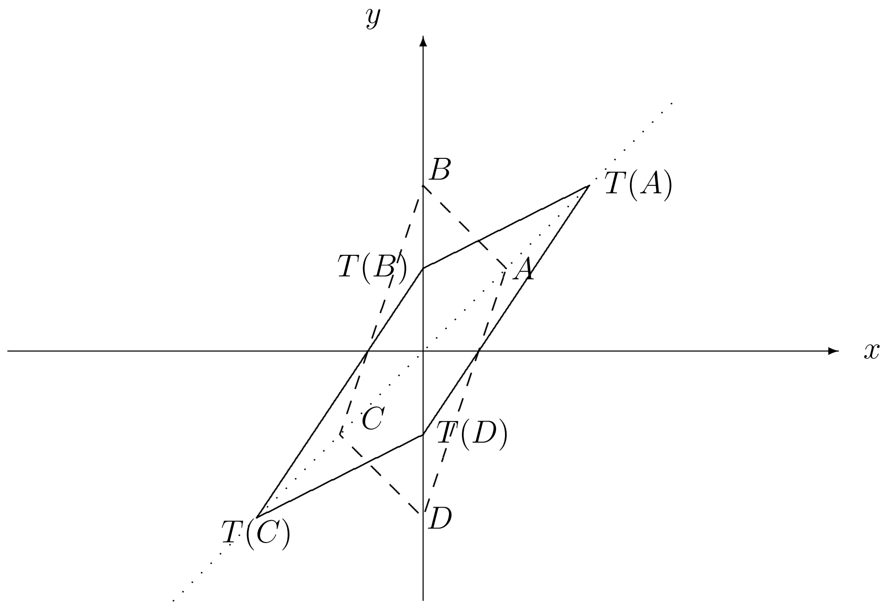
$$\boldsymbol{A}\begin{pmatrix}1\\-i\\0\end{pmatrix}=\begin{pmatrix}0&-1&0\\1&0&0\\0&0&-1\end{pmatrix}\begin{pmatrix}1\\-i\\0\end{pmatrix}=\begin{pmatrix}i\\1\\0\end{pmatrix}=i\begin{pmatrix}1\\-i\\0\end{pmatrix}$$

So $\begin{pmatrix}1\\-i\\0\end{pmatrix}$ is an eigenvector with eigenvalue i.

7. (i) Let $\boldsymbol{A}$ be the standard matrix of T.
The point $(1,1)$ is on the line $y=x$. So $T\begin{pmatrix}1\\1\end{pmatrix}=\boldsymbol{A}\begin{pmatrix}1\\1\end{pmatrix}=\begin{pmatrix}2\\2\end{pmatrix}$.
The point $(0,1)$ is on the y-axis. So $T\begin{pmatrix}0\\1\end{pmatrix}=\boldsymbol{A}\begin{pmatrix}0\\1\end{pmatrix}=\begin{pmatrix}0\\\frac{1}{2}\end{pmatrix}$.
By stacking, we have:

$$\boldsymbol{A}\begin{pmatrix}1&0\\1&1\end{pmatrix}=\begin{pmatrix}2&0\\2&\frac{1}{2}\end{pmatrix}\Rightarrow\boldsymbol{A}=\begin{pmatrix}2&0\\2&\frac{1}{2}\end{pmatrix}\begin{pmatrix}1&0\\1&1\end{pmatrix}^{-1}=\begin{pmatrix}2&0\\\frac{3}{2}&\frac{1}{2}\end{pmatrix}$$

(ii) A is on the line $x=y$, so $T(A)=T\begin{pmatrix}2\\2\end{pmatrix}=\begin{pmatrix}4\\4\end{pmatrix}$.
B is on the y-axis, so $T(B)=T\begin{pmatrix}0\\4\end{pmatrix}=\begin{pmatrix}0\\2\end{pmatrix}$.
C is on the line $x=y$, so $T(C)=T\begin{pmatrix}-2\\-2\end{pmatrix}=\begin{pmatrix}-4\\-4\end{pmatrix}$.
D is on the y-axis, so $T(D)=T\begin{pmatrix}0\\-4\end{pmatrix}=\begin{pmatrix}0\\-2\end{pmatrix}$.
The image of the parallelogram $ABCD$ (dashed lines) is given by the parallelogram $T(A)T(B)T(C)T(D)$ (solid lines) in the diagram below:



(iii) $\boldsymbol{A}\boldsymbol{u} = \begin{pmatrix} 8 \\ -4 \end{pmatrix} \Rightarrow \boldsymbol{u} = \boldsymbol{A}^{-1}\begin{pmatrix} 8 \\ -4 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ -\frac{3}{2} & 2 \end{pmatrix}\begin{pmatrix} 8 \\ -4 \end{pmatrix} = \begin{pmatrix} 4 \\ -20 \end{pmatrix}$.

(iv) $\boldsymbol{A}^3\begin{pmatrix} 16 \\ 0 \end{pmatrix} = \boldsymbol{A}^2\begin{pmatrix} 2 & 0 \\ \frac{3}{2} & \frac{1}{2} \end{pmatrix}\begin{pmatrix} 16 \\ 0 \end{pmatrix} = \boldsymbol{A}^2\begin{pmatrix} 32 \\ 24 \end{pmatrix} = \boldsymbol{A}\begin{pmatrix} 2 & 0 \\ \frac{3}{2} & \frac{1}{2} \end{pmatrix}\begin{pmatrix} 32 \\ 24 \end{pmatrix}$
$= \boldsymbol{A}\begin{pmatrix} 64 \\ 60 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ \frac{3}{2} & \frac{1}{2} \end{pmatrix}\begin{pmatrix} 64 \\ 60 \end{pmatrix} = \begin{pmatrix} 128 \\ 126 \end{pmatrix}$.

8. (i)

$$\begin{array}{rcrcr} j_{n+1} & = & & & 1.6a_n \\ a_{n+1} & = & 0.3j_n & + & 0.8a_n \end{array} \Longrightarrow \begin{pmatrix} j_{n+1} \\ a_{n+1} \end{pmatrix} = \begin{pmatrix} 0 & 1.6 \\ 0.3 & 0.8 \end{pmatrix}\begin{pmatrix} j_n \\ a_n \end{pmatrix}$$

(ii) $\boldsymbol{A}^{-1} = \dfrac{1}{-0.48}\begin{pmatrix} 0.8 & -1.6 \\ -0.3 & 0 \end{pmatrix}$

(iii) $\boldsymbol{x}_{2019} = \begin{pmatrix} 480 \\ 360 \end{pmatrix}$.
$\boldsymbol{x}_{2020} = \begin{pmatrix} 0 & 1.6 \\ 0.3 & 0.8 \end{pmatrix}\begin{pmatrix} 480 \\ 360 \end{pmatrix} = \begin{pmatrix} 576 \\ 432 \end{pmatrix}$.
So $j_{2020} = 576, \quad a_{2020} = 432$.
$\boldsymbol{x}_{2018} = \dfrac{1}{-0.48}\begin{pmatrix} 0.8 & -1.6 \\ -0.3 & 0 \end{pmatrix}\begin{pmatrix} 480 \\ 360 \end{pmatrix} = \begin{pmatrix} 400 \\ 300 \end{pmatrix}$.
So $j_{2018} = 400, \quad a_{2018} = 300$.

(iv) $\dfrac{j_{2018}}{a_{2018}} = \dfrac{400}{300} = \dfrac{4}{3}$.
$\dfrac{j_{2019}}{a_{2019}} = \dfrac{480}{360} = \dfrac{4}{3}$.

$\dfrac{j_{2020}}{a_{2020}} = \dfrac{576}{432} = \dfrac{4}{3}.$

The ratios are all equal to $\dfrac{4}{3}$.

$$\boldsymbol{A}\begin{pmatrix}4\\3\end{pmatrix} = \begin{pmatrix}0 & 1.6\\0.3 & 0.8\end{pmatrix}\begin{pmatrix}4\\3\end{pmatrix} = \begin{pmatrix}4.8\\3.6\end{pmatrix} = 1.2\begin{pmatrix}4\\3\end{pmatrix}$$

So $\begin{pmatrix}4\\3\end{pmatrix}$ is an eigenvector of $\boldsymbol{A}$.

Hence $\boldsymbol{x}_{2018}, \boldsymbol{x}_{2019}, \boldsymbol{x}_{2020}$ are all eigenvectors of $\boldsymbol{A}$, which are scalar multiples of $\begin{pmatrix}4\\3\end{pmatrix}$.

This explains why the ratios $\dfrac{j_n}{a_n}$ are all the same and equal to $\dfrac{4}{3}$.

9. (i) We have

$$\begin{aligned} a_{n+1} &= 0.9a_n + 0.05b_n \\ b_{n+1} &= 0.1a_n + 0.95b_n \end{aligned}$$

So $\boldsymbol{A} = \begin{pmatrix}0.9 & 0.05\\0.1 & 0.95\end{pmatrix}$.

(ii) $\boldsymbol{A}$ is diagonalizable with diagonalizing matrix $\boldsymbol{P} = \begin{pmatrix}1 & 1\\2 & -1\end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix}1 & 0\\0 & 0.85\end{pmatrix}$, we have

$$\begin{pmatrix}a_n\\b_n\end{pmatrix} = \boldsymbol{A}^n\begin{pmatrix}a_0\\b_0\end{pmatrix} = \boldsymbol{P}\boldsymbol{D}^n\boldsymbol{P}^{-1}\begin{pmatrix}a_0\\b_0\end{pmatrix} = \begin{pmatrix}1 & 1\\2 & -1\end{pmatrix}\begin{pmatrix}1 & 0\\0 & 0.85^n\end{pmatrix}\frac{1}{3}\begin{pmatrix}1 & 1\\2 & -1\end{pmatrix}\begin{pmatrix}600,000\\300,000\end{pmatrix}$$

In the long term, $0.85^n \to 0$ and

$$\begin{pmatrix}a_n\\b_n\end{pmatrix} \to \frac{1}{3}\begin{pmatrix}1 & 1\\2 & 2\end{pmatrix}\begin{pmatrix}600,000\\300,000\end{pmatrix} = \begin{pmatrix}300,000\\600,000\end{pmatrix}.$$

In the long term, the city will have a population of 300,000, and the suburbs will have a population of 600,000.

10. (i) $\begin{aligned} h_n &= 0.95h_{n-1} + 0.45s_{n-1} \\ s_n &= 0.05h_{n-1} + 0.55s_{n-1} \end{aligned}$,

which gives the matrix form $\boldsymbol{x}_n = \boldsymbol{A}\boldsymbol{x}_{n-1}$ with $\boldsymbol{A} = \begin{pmatrix}0.95 & 0.45\\0.05 & 0.55\end{pmatrix}$.

(ii) Suppose $\boldsymbol{x}_1 = \begin{pmatrix}80\\20\end{pmatrix}$. Then on the following day,

$$\boldsymbol{x}_2 = \boldsymbol{A}\boldsymbol{x}_1 = \begin{pmatrix}0.95 & 0.45\\0.05 & 0.55\end{pmatrix}\begin{pmatrix}80\\20\end{pmatrix} = \begin{pmatrix}85\\15\end{pmatrix}$$

So the percentage of students likely to be sick: $15/(85+15) = 15\%$.

(iii) $\boldsymbol{A}$ is diagonalizable with diagonalizing matrix $\boldsymbol{P} = \begin{pmatrix} -1 & 9 \\ 1 & 1 \end{pmatrix}$ and $\boldsymbol{D} = \begin{pmatrix} 0.5 & 0 \\ 0 & 1 \end{pmatrix}$.

So $\boldsymbol{A} = \begin{pmatrix} -1 & 9 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0.5 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 9 \\ 1 & 1 \end{pmatrix}^{-1}$.

For large n, $\boldsymbol{A}^n = \begin{pmatrix} -1 & 9 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 9 \\ 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 0.9 & 0.9 \\ 0.1 & 0.1 \end{pmatrix}$.

Then $\boldsymbol{A}^n \begin{pmatrix} h \\ s \end{pmatrix} = \begin{pmatrix} 0.9 & 0.9 \\ 0.1 & 0.1 \end{pmatrix} \begin{pmatrix} h \\ s \end{pmatrix} = \begin{pmatrix} 0.9(h+s) \\ 0.1(h+s) \end{pmatrix}$.

So 1 in 10 students (or 10%) will be sick at steady state.